

Final Project: Company RAG Chatbot with n8n

In this project, each group develops a simple but functional RAG chatbot for a fictional company.

The technical goal is not to build a complex application, but to create a working chatbot that can answer company-specific questions based on an internal document database.

Each group receives one company scenario. For this scenario, the group must create or prepare at least five internal company documents. Each document should have at least one page of content and be uploaded as PDF.

Group 1 – Case 1

Group 2 – Case 2

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The chatbot must be able to answer at least three relevant business questions based on these documents.

The final result is a working n8n workflow, a short documentation and a final presentation.

Technical Requirements

- The chatbot must be implemented in n8n.
- Recommended setup: n8n + Supabase + OpenRouter/OpenAI or similar APIs.
- The workflow must be submitted as a JSON export.
- The chatbot must function during the presentation.

Minimum Requirements

- Working n8n RAG chatbot as JSON export
- Three example questions the chatbot can answer
- Documentation (max. 10 pages)
- 15-minute presentation
- 10-minute Q&A session

Use of AI Tools

- AI tools may be used freely for document generation, workflow creation, prompting, debugging and presentation preparation.
- The important part is understanding and explaining the solution clearly.

Documentation

- Maximum 10 pages excluding appendix.
- Should include company situation, architecture, workflow, example questions, risks, EU AI Act analysis and implementation reflection.
- The documentation is supporting material. The main grading focus is the presentation.

Presentation

- 15 minutes presentation + 10 minutes Q&A.
- Must include workflow explanation, chatbot demo, architecture, benefits, risks and safeguards.

Submission

Submit the following documents under the assignment tab in Teams until 31.05:

- Project Documentation (one per group)
- Presentation (one per group)
- Individual Grading report (everyone)

Case 1: HR Knowledge Assistant

Company Situation

BergTech Maschinenbau GmbH is a medium-sized manufacturing company with 450 employees. The HR department receives many repetitive questions regarding onboarding, vacation rules, training procedures and internal HR processes.

Required Document Database

- Onboarding Guide
- Vacation and Absence Policy
- Training and Compliance Policy
- HR FAQ Document
- Employee Offboarding Checklist

Example Questions

1. What documents does a new employee need to submit before their first working day?
2. How many days in advance should vacation be requested?
3. Which mandatory trainings must new employees complete during onboarding?

Main Focus

Employee knowledge access, onboarding efficiency and HR compliance.

Important Risks

Incorrect HR guidance, hallucinated policy information and processing of sensitive employee data.

Case 2: Technical Maintenance Assistant

Company Situation

Vektor Industrial Systems GmbH produces industrial packaging machines. Technicians struggle to quickly find troubleshooting and maintenance information during downtime situations.

Required Document Database

- Machine Maintenance Manual
- Troubleshooting Guide
- Safety Instructions
- Spare Parts Overview
- Preventive Maintenance Checklist

Example Questions

1. What should a technician check first if the conveyor belt stops unexpectedly?
2. Which safety steps are required before opening the machine housing?
3. How often should the main drive unit be inspected?

Main Focus

Technical knowledge retrieval, troubleshooting efficiency and operational support.

Important Risks

Incorrect technical instructions could cause safety issues or production downtime.

Case 3: Compliance & Policy Assistant

Company Situation

Nova Insurance Group faces frequent employee questions regarding data privacy, AI policies and compliance procedures.

Required Document Database

- Internal AI Usage Policy
- Data Privacy Guideline
- Customer Data Handling Policy
- Compliance Approval Process
- EU AI Act Internal Briefing

Example Questions

1. May employees enter customer data into public AI tools?
2. When does a new AI use case require compliance approval?
3. What must be documented when AI-generated content is used internally?

Main Focus

Governance, compliance support and AI policy awareness.

Important Risks

False compliance guidance and legal liability.

Case 4: Customer Support Knowledge Assistant

Company Situation

UrbanHome E-Commerce GmbH handles many repetitive customer support requests regarding delivery, warranty and returns.

Required Document Database

- Return Policy
- Delivery and Shipping FAQ
- Warranty Guidelines
- Product Information Sheet
- Complaint Handling Process

Example Questions

1. Under what conditions can a customer return a damaged product?
2. What delivery time should be communicated for large furniture items?
3. When should a customer complaint be escalated to a supervisor?

Main Focus

Customer support efficiency and faster information retrieval.

Important Risks

Incorrect customer information and hallucinated responses.

Case 5: University Student Services Assistant

Company Situation

Hohenberg University of Applied Sciences supports around 6,000 students. The student services office receives many repetitive questions about enrollment, exams, deadlines, course registration and study regulations.

Required Document Database

- Enrollment Guide
- Examination Regulations
- Course Registration FAQ
- Student Services Process Guide
- Academic Calendar and Deadline Overview

Example Questions

- What documents are required for enrollment?
- Until when can students register for exams?
- What should a student do if they miss an important deadline?

Main Focus

Student self-service, faster access to administrative information and support for university staff.

Important Risks

Incorrect academic guidance, outdated deadline information and students relying on wrong information for exams or enrollment.